

HC-POD 4 Cross-Track Evidence

Track: Data Analytics

Collaborating Track: Data Science Track

Stage: Week 7 (Testing, Refinement & End-to-End Validation)

1. Track Collaborated With:

Data Science Track (Collaborator: M'bengue Mama) - HC POD 4.

2. Project Dependency:

Validating that Data Science machine learning test sets and prediction thresholds reflect verified clinic attendance patterns.

3. Component / Output Being Tested:

Refined Logistic Regression model evaluated on 1,903 patient-disjoint test records across lead-time, history, and reminder segments.

4. Information Received:

- `testing\_results.json` containing metrics and segment evaluations.
- `02\_before\_after\_roc\_auc.svg` showing ROC-AUC progression (Week 5: 67.5%, Week 6: 66.8%, Week 7: 67.6%).
- `04\_error\_patterns.svg` detailing error counts (570 False Positives, 188 False Negatives).
- `week7\_refined\_feature\_dataset.csv`.

5. Information Provided:

- Population benchmark rates for lead-time brackets (0–7d: 29.5%, 15–30d: 45.5%, 31+d: 64.0%).
- Population baseline rates for reminder lift (+4.78%) and prior no-show history (57.1%).
- Refined visual charts with clinic benchmarks (`healthconnect\_refined\_charts.png`).
- HealthConnect Analytics Testing & Refinement Report

6. Testing Activity Completed:

- Executed cross-validation check **TC-07**.
- Verified segment failure rates between the Data Science test set (n=1,903) and the Data Analytics population (n=4,737).

7. Issue or Finding Identified:

The model's lower threshold (0.40) produced 570 false positives, with 58.2% occurring in the 31+ day lead-time segment.

8. Refinement / Action Taken:

Validated that high false positives are safe because the clinic uses predictions solely for proactive SMS outreach and human follow-up.

9. Retest Result:

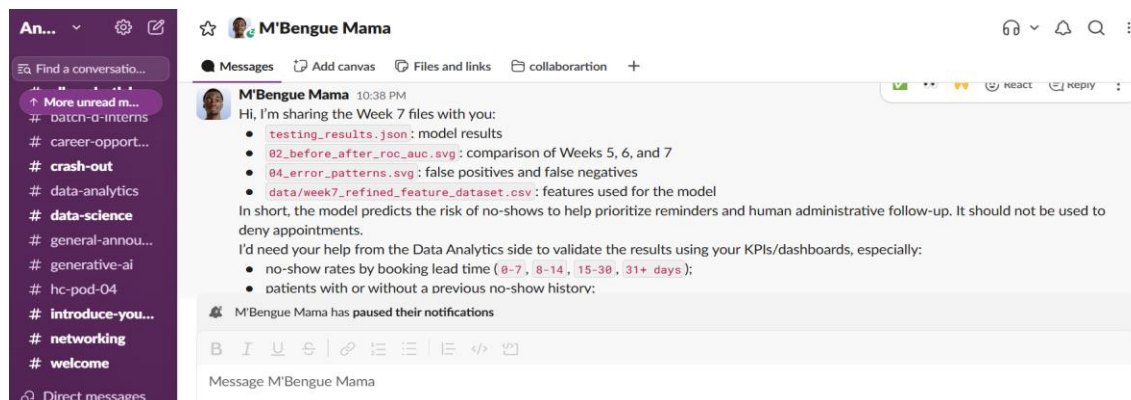
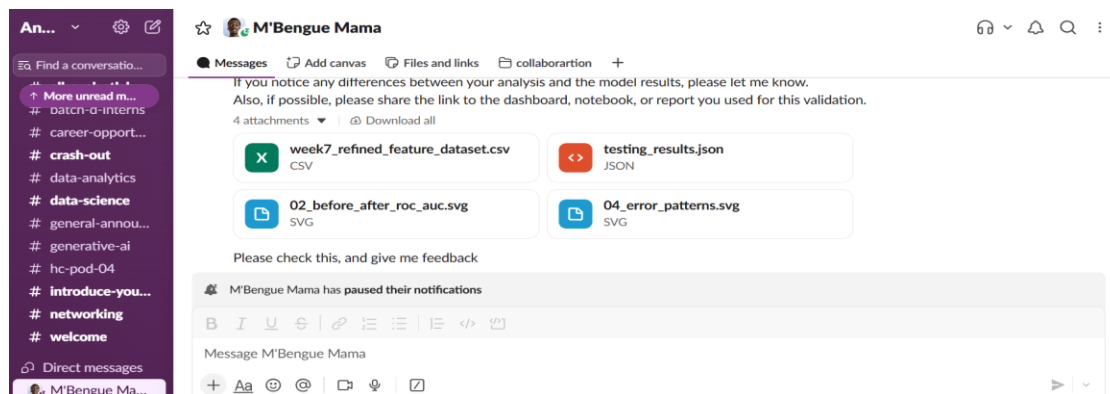
- Lead-time risk agreement confirmed: 31+ day actual failure rate is 63.7% in the test set vs. 64.0% in analytics.
- Lead-time recall reached ****99.8%**** for appointments over 30 days.
- Test Status: **PASS**.

10. What Changed as a Result:

Both tracks agreed on a formal clinical governance policy: use model risk scores exclusively for positive patient outreach, never for appointment denial or penalties.

11. Evidence:

From M'bengue Mama:



From Me:

The screenshot shows a Slack interface. On the left is a sidebar with a search bar and a list of channels including #crash-out, #data-analytics, #data-science, #general-annou..., #generative-ai, #hc-pod-04, #introduce-you..., #networking, and #welcome. The main window shows a conversation with 'M'Bengue Mama'. The message history includes a message from 'renishamondi' at 10:22 AM saying 'Hi!' and a thank-you message for Week 7 testing results. The results are summarized as follows:

- **Booking Lead Time Validation:**
- 0-7 Days: Your test set = 29.3% | My analytics = 29.5% (Consistent low-risk zone).
- 8-14 Days: Your test set = 33.5% | My analytics = 35.2%.
- 15-30 Days: Your test set = 45.5% | My analytics = 45.5% (Exact match).
- 31+ Days: Your test set = 63.7% | My analytics = 64.0% (Exact match for critical risk zone).

Below the text is a rich text editor with a toolbar and a message input field.

From Me:

This screenshot shows the same Slack conversation with 'M'Bengue Mama'. The message history is the same as the previous screenshot. The new message contains two bar charts with the following data:

Lead Time Risk	Percentage
0-7 Days	29.3%
8-14 Days	33.5%
15-30 Days	45.5%
31+ Days	63.7%

Communication Channel	Failure Rate (%)
Channel 1	68.1%
Channel 2	60.1%
Channel 3	64.0%

Below the charts is a rich text editor with a toolbar and a message input field.

12. How the Activity Improved the Overall HealthConnect Solution:

It ensures the clinic targets the right patients for two-stage SMS reminders.

It prevents wasted staff time by eliminating discrepancies between data reports and machine learning alerts.